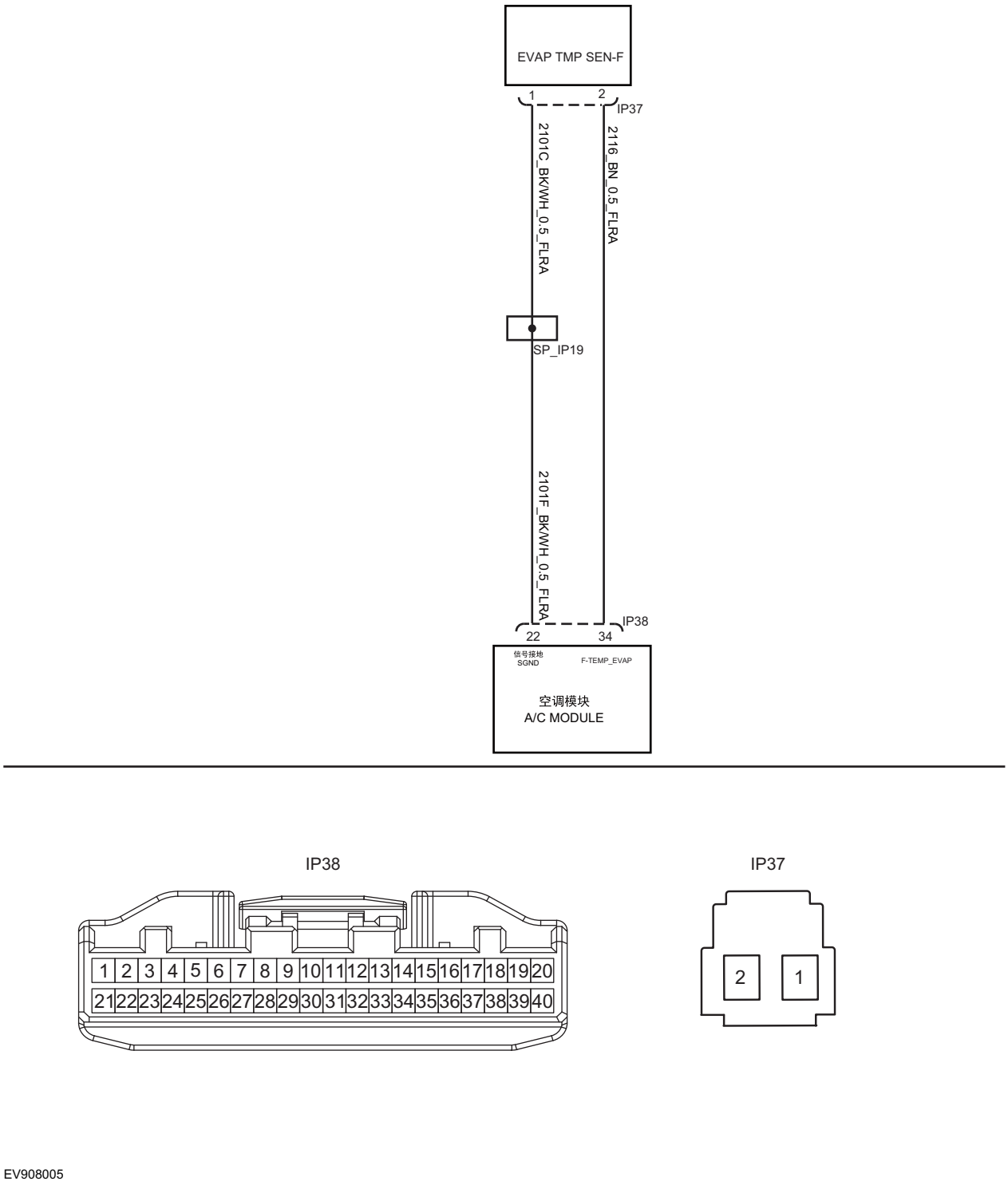
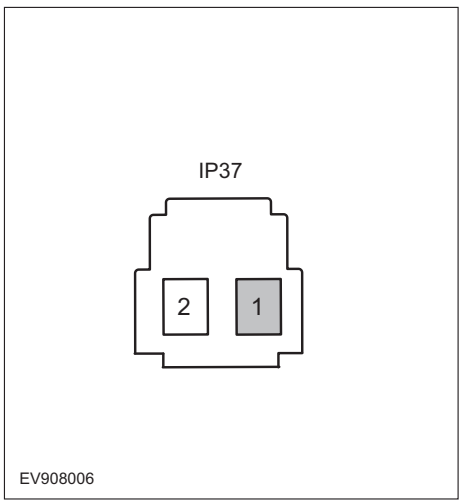
B1602-1C

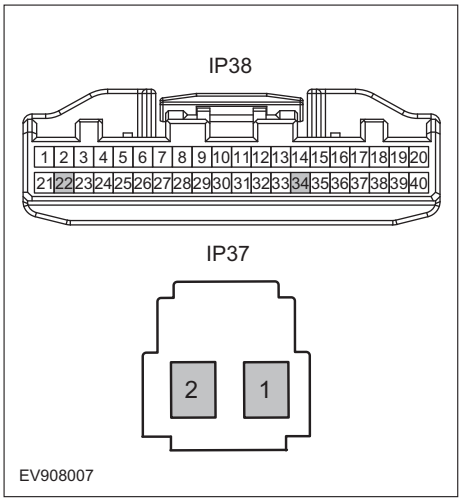
Description of fault diagnosis codes
B1602-1C：Evaporator sensor short or open
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • Air conditioning controller fault • Temperature sensor of the evaporator fault
Conditions for fault code operation
The ignition switch is in ON position.
Conditions for fault code setting
B1602-1C
To set the effect of a fault code condition
1 2
Description of circuit diagram
The air conditioning control module monitors the evaporator temperature sensor via low level reference circuit and signal circuit. The air conditioning control module detects the voltage value inversely proportional to temperature on the sensor. When the ambient temperature is low, the sensor resistance will be high. When the ambient temperature is high, the sensor resistance will be low.

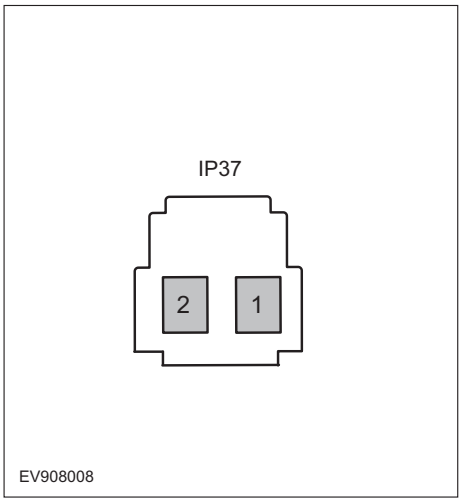
Circuit diagram

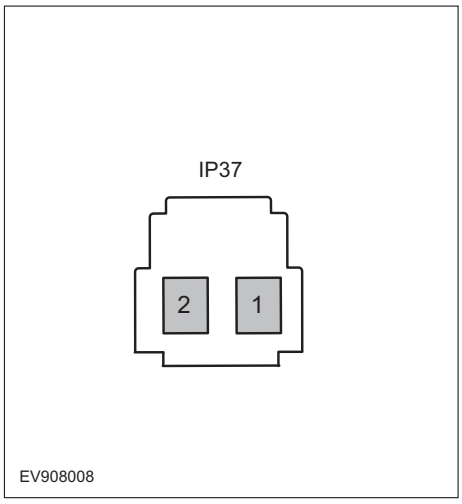


EV908005

Troubleshooting procedures		
Step 1	General inspection.	
		1 Check the fuse for damage. 2 Check battery capacity. 3 Check the evaporator temperature control module and the harness plug of air conditioning control module for damage, poor contact, aging, and looseness. 4 Check whether the result is normal or not?
		NO ➞ Repair or replace the faulted parts.
YES ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the air condition control system with diagnostic apparatus. 4 Is there any other fault code except for B1602-1C?
		YES ➞ Refer to:DTC Summary List (HVAC).
NO ↓		
Step 3	Check the power supply circuit of evaporator temperature sensor.	
		1 Turn the ignition switch to OFF. 2 Disconnect the harness plug IP37 of evaporator temperature sensor. 3 Turn the ignition switch to ON. 4 Measure the voltage between evaporator temperature sensor and ground. Measure circuit: voltage between terminal 2 on plug IP37 and ground. **Standard value:4.8-5.2V** 5 Check whether the result is normal or not?
		NO ➞ Please repair or replace the faulted parts.
YES ↓		

Step 4	Check the power supply circuit of evaporator temperature sensor(open circuit).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug IP38 of air conditioning controller. 3 Measure the resistance of evaporator temperature sensor harness terminal and air conditioning controller harness terminal. Measuring circuit: resistance between terminal 1 on plug IP37 and terminal 22 on plug IP38. Measuring circuit: resistance between terminal 2 on plug IP37 and terminal 34 on plug IP38. **Standard value: ≈ 0Ω** 4 Check whether the result is normal or not?
	NO → Please repair or replace the faulted parts.
YES ↓	

Step 5	Check the power supply circuit of evaporator temperature sensor(short circuit).
	1 Turn the ignition switch to OFF. 2 Measure the resistance between the harness plug of evaporator temperature sensor and ground. Measuring circuit: resistance between terminal 1 on plug IP37 and ground. Measuring circuit: resistance between terminal 2 on plug IP37 and ground. **Standard value: ∞** 3 Measure the voltage between the harness plug of evaporator temperature sensor and ground. Measure circuit: voltage between terminal 1 on plug IP37 and ground. Measure circuit: voltage between terminal 2 on plug IP37 and ground. **Standard value: ≈ 0V** 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 6	Check the evaporator temperature sensor(components).
	1 Measure the resistance of terminals on evaporator temperature sensor directly. Measuring circuit: measure the resistance between terminal 1 and 2 on plug IP37. **Standard value:** 2 Measure the resistance between the evaporator temperature sensor terminal and the housing. Measuring circuit: Measure the resistance between the terminal 1 on plug IP37 and the housing. Measuring circuit: Measure the resistance between the terminal 2 on plug IP37 and the housing. **Standard value: ∞** 3 Check whether the result is normal or not?
	NO → Replace the ambient temperature sensor.
YES ↓	
Step 7	Check if air conditioning module is normal.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug IP38 of air conditioning controller. 3 Check and repair; • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the air conditioning control module plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Replace the air conditioning control module.
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems,	
Next Step ↓	
Diagnosis end.	